

David Kivlichan

Machine Learning Engineer - 5+ years

Experience

- Oct 2025 – **Senior Machine Learning Engineer**, Optro (Formerly AuditBoard), AI & ML.
Present
- Refactored the machine learning evaluation service into a standalone, production-grade system, improving reliability, scalability, and maintainability.
 - Integrated large language model evaluation pipelines into continuous integration (CI), enabling automated regression and quality checks for LLM-powered features.
 - Provisioned and deployed new ML infrastructure using Terraform and Terragrunt, including managed LLM (GPT-5.2) and Azure Speech Services resources.
- Nov 2022 – **Senior Applied Scientist**, RailVision Analytics, Machine Learning & MLOps.
Oct 2025
- Achieved 10–20%+ fuel savings with \$XM+ ARR impact by leading development of a fuel-optimization recommendation system for freight and passenger rail.
 - Designed and deployed scalable ML models for fuel optimization using Python, PyTorch, and scikit-learn on AWS (S3, RDS, EC2).
 - Led the end-to-end ML lifecycle, integrating Airflow, MLFlow, and Docker to automate training, validation, and deployment workflows.
 - Optimized inference pipelines to reduce compute costs and latency while serving real-time recommendations.
 - Contributed to a physics-informed simulator with $< 1\%$ RMSE to validate model predictions and ensure safety compliance.
- Jan 2022 – **Applied Scientist**, Yelp Inc, Machine Learning & MLOps.
Nov 2022
- Generated \$1M+ predicted ARR by deploying ads budget pricing machine learning models at scale using PySpark, TensorFlow, and Kubernetes to optimize advertiser engagement and budget allocation.
 - Researched and implemented predictive models for customer lifetime value estimation, improving targeting strategies for sales and marketing.
- 2019 – 2021 **Graduate Student AI Researcher**, Royal Bank of Canada & University of Toronto, Deep Learning & Security.
- Designed adversarial deep learning techniques for malware detection, reducing false positives by 12% at fixed recall.
 - Conducted applied research at Borealis AI and RBC's Joint Security Operations Center on NLP-based cybersecurity systems.
- 2017 – 2018 **Product Engineering Intern**, Microchip, Data Science & Optimization.
- Developed predictive models for silicon wafer selection, improving product yield by \$1M+ per month.

Education

- 2019 – 2021 **MASc in Industrial Engineering**, *University of Toronto*, Thesis: Deep learning applications in cybersecurity, GPA: 4.0.
Teaching Assistant for Engineering Design, Dynamics, and Differential Equations.
- 2014 – 2019 **Honours BSc in Engineering Science (Physics)**, *University of Toronto*, percent average: 83.5%, GPA: 3.7/4.0.

Skills

- MLE Deep Learning, PyTorch, TensorFlow, scikit-learn, NLP, Recommender Systems, Time-Series Analysis
- ML Systems Airflow, MLFlow, Docker, Kubernetes, AWS (S3, RDS, EC2, Lambda), Redis, InfluxDB
- SWE Python, PySpark, SQL, Git, Unix Systems, CI/CD, API Design